SC7 Bayes Methods MT26

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Lecture 2: a first case study in Bayesian inference

Case study: Radiocarbon dating

The data come from the site of an ancient settlement in NZ.

Archaeologists dug down through a habitation band. Above and below this layer there is no sign of activity. They dated $n = 7$ charcoal samples from the habitation layer.

The researchers are very confident (before seeing the data) that

- there was no settlement before $U = 1000$ years BP;
- the settlement had been abandoned prior to $L = 500$ BP.

Key scientific question: For how long was the site settled?

There is a suggestion that this camp was settled for just weeks or months rather than years.

This is a question about the occupation span.

Observation model

Uncalibrated radiocarbon date $y_i,\ i = 1, 2, ..., n$ is a biased measurement of true age θ_i of specimen i. Observation model:

$$y_i = \mu(\theta_i) + \epsilon_i$$

with independent errors

$$\epsilon_i \sim N(0, \sigma_c(\theta_i)^2 + \sigma_i^2).$$

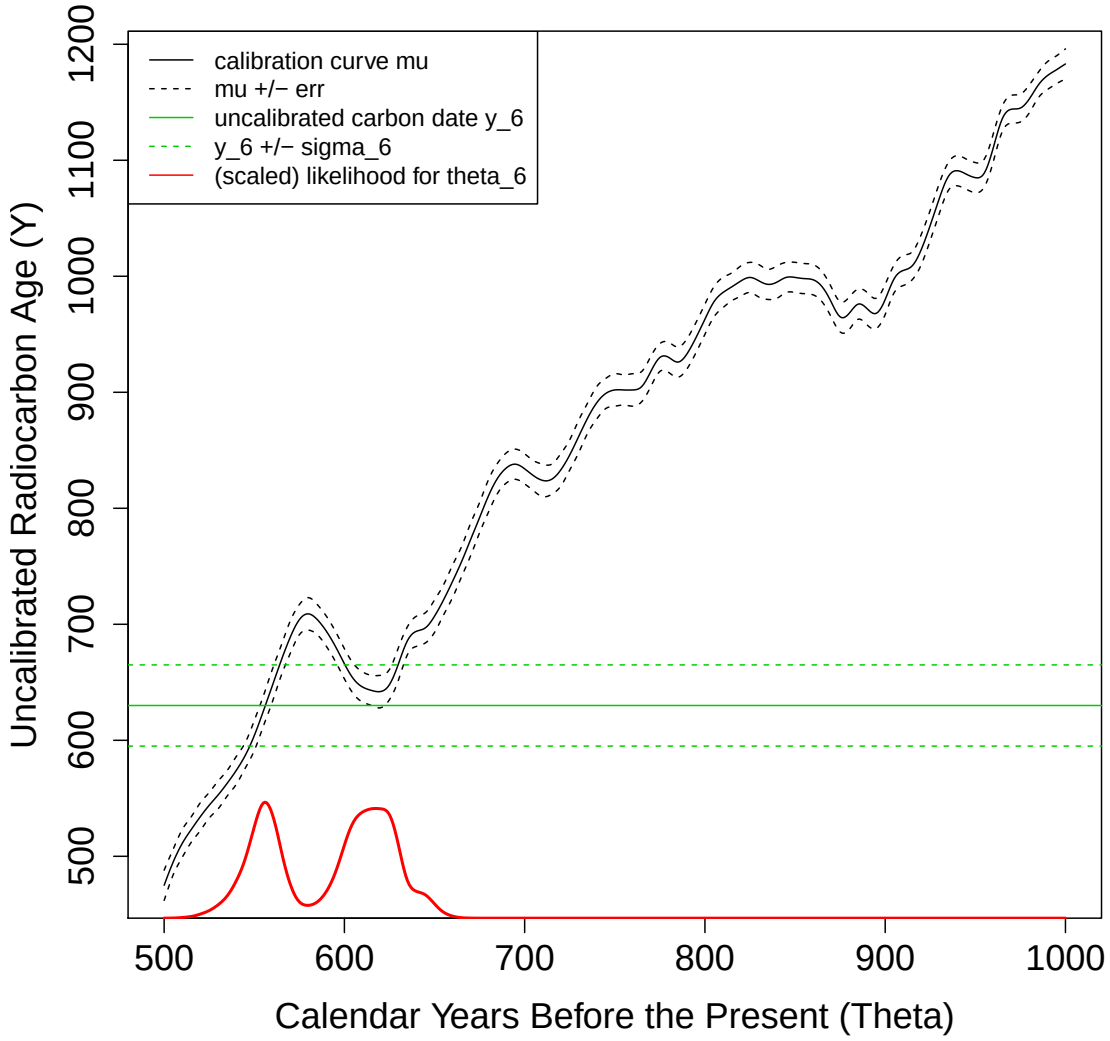
Two sources of noise: measurement error σ_i and uncertainty $\sigma_c(t)$ in $\mu(t)$. Calibration functions $\mu(t)$ and $\sigma_c(t)$ are known.

The likelihood factor for one unknown age parameter θ_i is:

$$p(y_i|\theta_i) = \frac{\exp(-(y_i - \mu(\theta_i))^2/2(\sigma_c(\theta_i)^2 + \sigma_i^2))}{\sqrt{2\pi(\sigma_c(\theta_i)^2 + \sigma_i^2)}}$$

Calibration curve μ with likelihood $p(y_6|\theta_6)$ for sixth parameter.

The functions $\mu(\theta)$ and $\sigma_c(\theta)$ are available from radiocarbon dating labs (Eg OxCal, a widely used Bayesian package for RCD).



The data

$n = 7$ radiocarbon dates.

```
#Date id - i - y_i - sigma_i
#NZ 7758 - 1 - 580 - 47
#NZ 7761 - 2 - 600 - 50
#NZ 7757 - 3 - 537 - 44
#NZ 7756 - 4 - 670 - 47
#NZ 7755 - 5 - 646 - 47
#WK 2589 - 6 - 630 - 35
#NZ 7771 - 7 - 660 - 46
```

The likelihood plot in the previous graph showed $p(y_6|\theta_6)$, where the overall likelihood is

$$p(y|\theta) = \prod_{i=1}^{n} p(y_i|\theta_i).$$

Priors

A simple uniform prior $\pi_u(\theta)$ for $\theta = (\theta_1, ..., \theta_n)$ might set lower and upper bounds $L = 500$ and $U = 1000$ on the true ages of the specimen, and assert that all other values are equally probable:

$$\pi_u(\theta) \propto \prod_i \mathbb{I}(L \leq \theta_i \leq U).$$

The parameters space is $\Omega_u = [L, U]^n$.

We could estimate the start and end of settlement using $\theta^- = \min(\theta)$ and $\theta^+ = \max(\theta)$ the most recent and oldest dates, and estimate the span using

$$S_u = \theta^+ - \theta^-.$$

.

Does this prior meet our elicitation criteria?

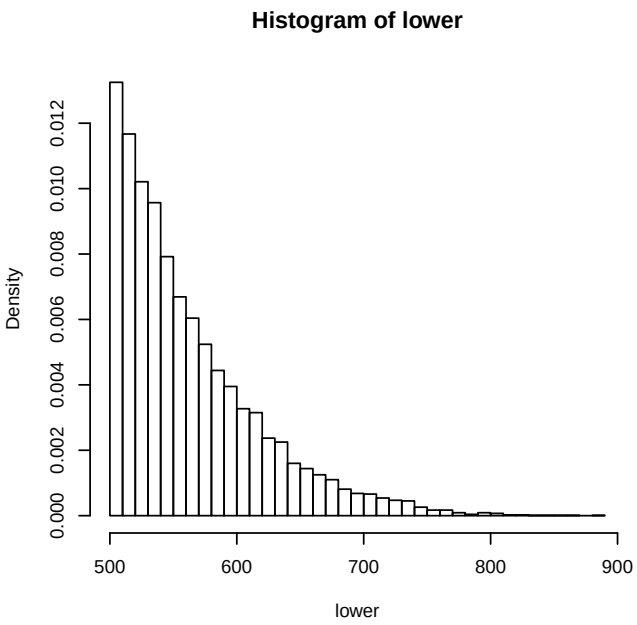
(A) can we think of the dates $\theta_i, i = 1, 2, ..., n$ as generated by some "physical" process?

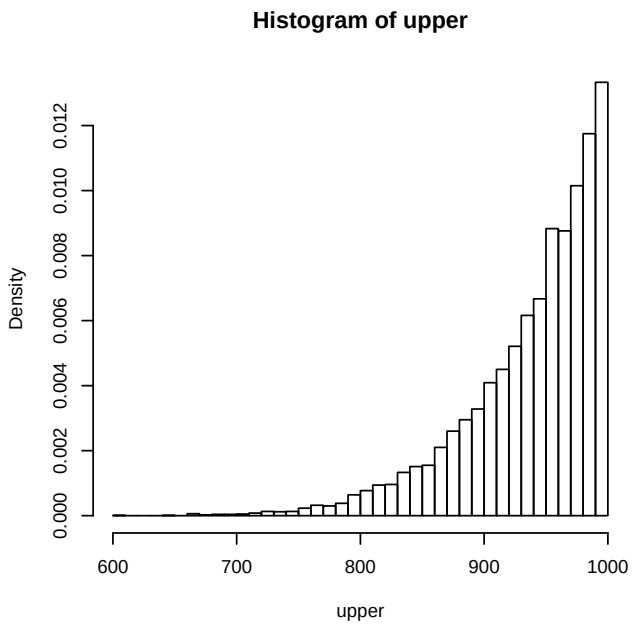
(B) the span is linked to a key scientific question - is the prior non-informative on this variable?

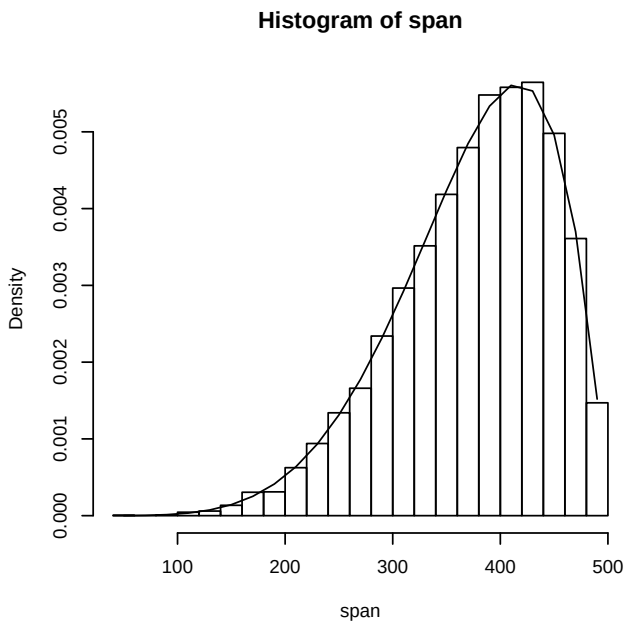
Does π_u meet these criteria? Certainly not (A). For (B) we need the marginal prior distribution $\pi_{S_u}(s)$ of the span S_u. This is

$$\pi_{S_u}(s) = \frac{n(n-1)}{(U-L)^n} s^{n-2}(U-L-s) \quad \text{for } 0 \leq s \leq U-L.$$

We dont actually *need* to calculate this as we can get the histogram by simulating the prior.







Marginal priors $\pi_u(\theta^-)$ (left), $\pi_u(\theta^+)$ (mid), $\pi_{S_u}(s)$ (right). We have no basis for favoring large span s values over smaller values.

When $s \ll U - L$ then $\pi_{S_u}(2s) \simeq \pi_{S_u}(s)2^{n-2}$ a strong bias for larger span. We are biasing the thing we want to estimate!

π_u was a bad choice as it does not represent prior knowledge.

A prior from a process generating θ

Model assumptions:

- Settlement starts at time $\psi_2 < U$ and ends at $L < \psi_1 < \psi_2$.
- The span is $S_s = \psi_2 - \psi_1$.
- Dates θ are a realisation of a Poisson process* with rate λ. We choose the number of dates so we condition on n.

Consequences:

- We have $\theta_i \sim U(\psi_1, \psi_2)$ iid (conditioning a Poisson process on the number of events gives uniform event times).
- Our prior for $\theta|\psi$ is

$$\pi_s(\theta|\psi) = \frac{1}{(\psi_2 - \psi_1)^n}\mathbb{I}(\psi_1 < \theta_1, ..., \theta_n < \psi_2).$$

*See notes Chapter 1 Appendix B for some background on the Poisson process

We have to specify the prior for $\psi = (\psi_1, \psi_2)$. We would like the marginal prior span to be uniform. The prior

$$\pi_s(\psi) \propto \frac{1}{U - L - (\psi_2 - \psi_1)}$$

has a uniform distribution[†] on values of $\psi_2 - \psi_1$ so this is non-informative with respect to the span.

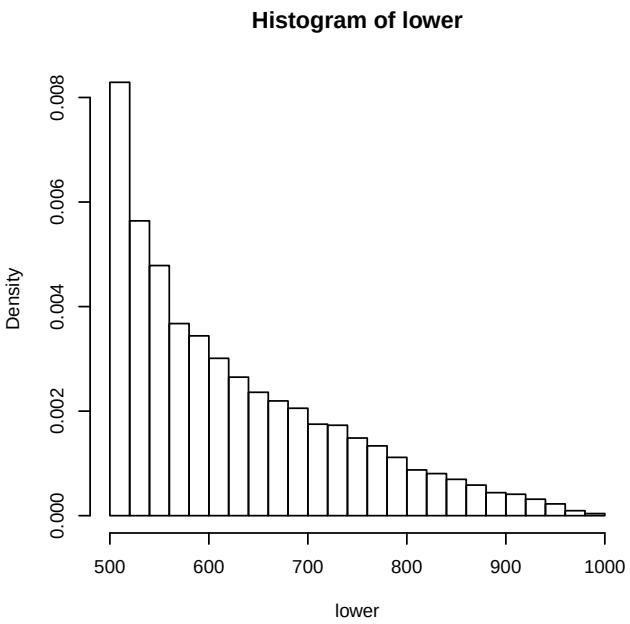
Since $\pi_s(\theta, \psi) = \pi_s(\theta|\psi)\pi_s(\psi)$,

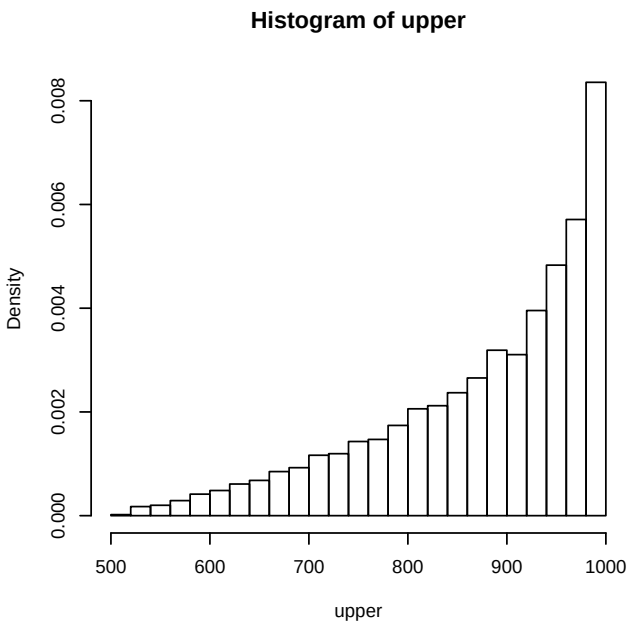
$$\pi_s(\theta, \psi) \propto \frac{1}{(\psi_2 - \psi_1)^n} \frac{1}{(U - L - (\psi_2 - \psi_1))}$$

models the prior information we actually have. The parameter space is

$$\Omega_s = \{(\theta, \psi) \in [L, U]^{n+2} : \psi_1 < \theta_i < \psi_2, i = 1, ..., n\}$$

[†]See notes exercises 1.3-1.5 and Chapter 1 Appendix A for proof







These marginal priors $\pi_s(\psi_1)$, $\pi_s(\psi_2)$ and $\pi_{S_s}(s)$ better represent the prior information we actually possess.

The prior on span is uniform, and the priors $\pi_s(\psi_1)$, $\pi_s(\psi_2)$ distribute probability mass more evenly over the parameter domain.

Radiocarbon example continued ... Sampling the posterior

We have two posterior distributions,

$$\pi_u(\theta|y) \propto p(y|\theta)\pi_u(\theta)$$

and

$$\pi_s(\theta, \psi|y) \propto p(y|\theta)\pi_s(\theta, \psi).$$

The two posterior distributions have the same likelihood, but different priors π_u and π_s, and parameters spaces Ω_u and Ω_s.

We would like to summarise these distributions, form histograms and HPD credible sets.

I used simple random walk Metropolis Hastings MCMC* updating one variable at a time. We look at MCMC later in the course.

I sampled $\theta^{(t)} \sim \pi_u(\theta|y), t = 1, ..., T$, computed the order statistics $\theta^{(t)}_{(1)}, \theta^{(t)}_{(n)}$ and span $S^{(t)}_u = \theta^{(t)}_{(n)} - \theta^{(t)}_{(1)}$ for each MCMC sample.

I also sampled $(\theta^{(t)}, \psi^{(t)}) \sim \pi_s(\theta, \psi|y), t = 1, ..., T$, in the second model, and computed the span $S^{(t)}_s = \psi^{(t)}_2 - \psi^{(t)}_1$ for each MCMC sample.

I plotted histograms of the samples $S^{(t)}_u$ and $S^{(t)}_s$, $t = 1, \ldots, T$ and used them to estimate HPD sets.

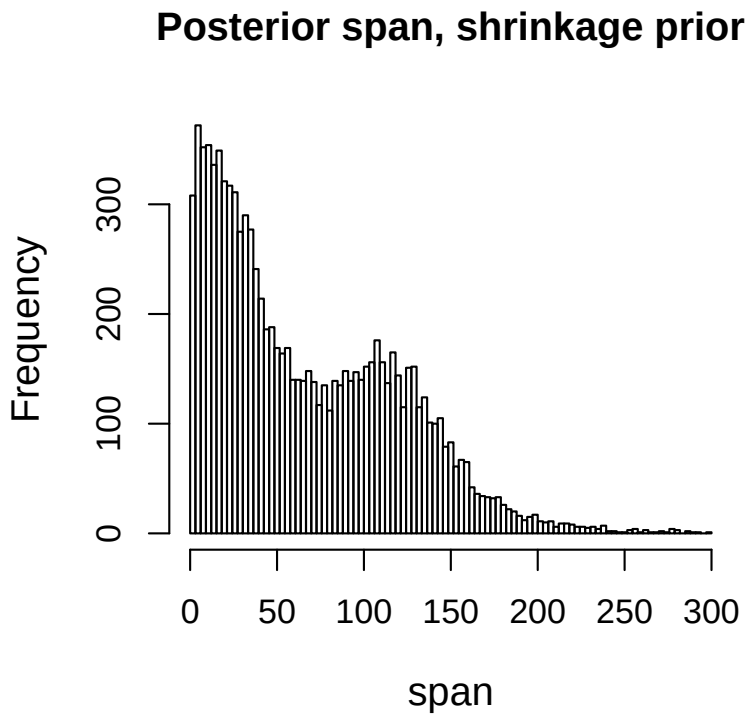
*See attached R-code for details.

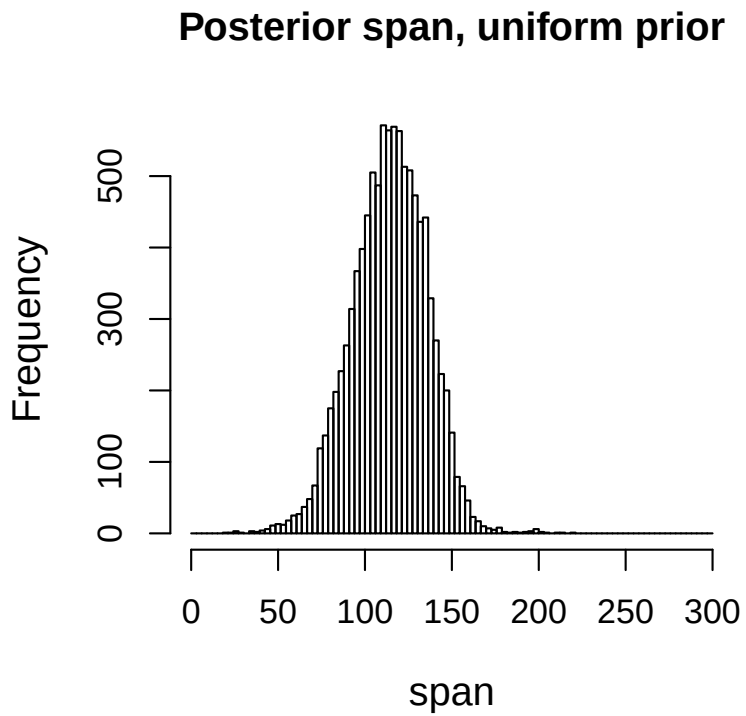
Summarising the results

We are interested in the span, and comparing two models with and without shrinkage. Here are the two HPD sets.

Model 2 (shrink/π_s): [0,160] Model 1 (unif/π_u): [70,160]

The marginal posterior histograms differ in shape and support.





Bayes Factor compares models. Which model fits the data best?

Under model one the marginal likelihood $p_u(y)$ say, is

$$p_u(y) = \int_{\Omega_u} p(y|\theta)\pi_u(\theta)d\theta$$

and under model two we have

$$p_s(y) = \int_{\Omega_s} p(y|\theta)\pi_s(\theta,\psi)d\psi d\theta$$

We estimate these using Monte Carlo.[†]

We get $\hat{p}_u \simeq 4 \times 10^{-21}$ and $\hat{p}_s \simeq 8 \times 10^{-19}$.

The Bayes factor $p_s(y)/p_u(y)$ is about $\hat{B}_{s,u} \simeq 200$.

The shrinkage prior π_s is clearly favoured over the uniform prior.

[†]See Section 6 of notes and lecture 8 for stable estimators.

Conclusions

We conclude that a brief settlement time is not ruled out (see histogram of S_s) in our favored analysis. It is ruled out by an analysis using a prior which weights against brief settlement times.

We dont have much data (7 noisy numbers) so the conclusions show some sensitivity to the choice of prior. However the prior π_s better represented actual prior knowledge than π_u and was strongly favored by the data ($B_{s,u} \simeq 200$).

The process-model based prior π_s allows very small spans close to zero, while the uniform prior rules them out. This is clearly a case where we dont want the prior to impose structure we cant support on prior grounds.